

s It is worth noting that there is no clear cut between what constitutes safe and unsafe queries, and no standard on how compliant the answer should be. Therefore, the added flexibility of adapting the model to different refusal standards for different usage scenarios is important.

Through partial orthogonalization, we can control the distance between the candidate false refusal and true refusal vectors by controlling the coefficient λ , as introduced in Eq. 8. By decreasing λ , the original false refusal vector is less modified by the orthogonalization, which leads to a stronger refusal removal when ablated. In Figure 2, as we decrease λ , model responses become more compliant. When we set λ to 1, the model refuses to give the instruction because it regards “cutting off the heads of fish” as *not safe* and *not hygienic*. As we lower λ to 0.8, the model starts to give a compliant answer but cautions against it as it is *harmful to the health of the fish* and *affects the flavor and texture*. With a lower λ value of 0.6, the model is fully compliant. Therefore, λ enables us to control the safety sensitivity of the model in a very fine-grained way.

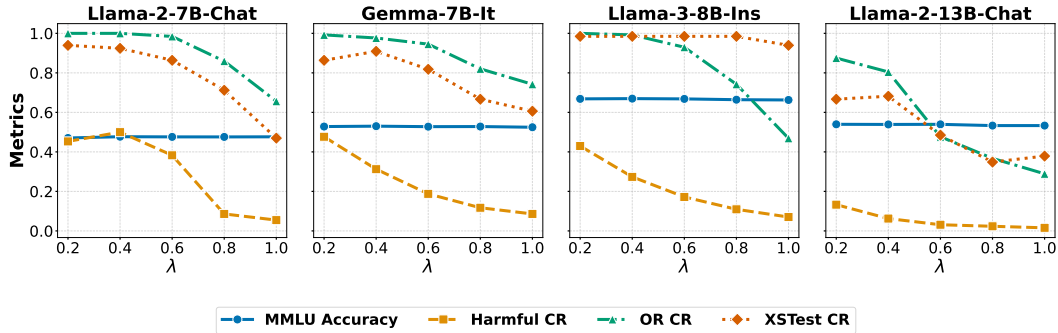


Figure 3: MMLU accuracy and compliance rate (CR) to pseudo-harmful (OR, XSTest) and harmful data. Changing the λ value can adjust the sensitivity to safety-related questions. Lowering λ can make the model less sensitive and more open to answering questions. The model’s general capability is unaffected since we adopt a surgical approach by only selecting vectors that have minimal effect on the output distribution.

To understand the impact of the λ value comprehensively, we plot in Figure 3 the MMLU performance and the compliance rate on OR-Bench-Hard (**OR CR**), XSTest-Safe Hard (**XSTest CR**) and harmful data D_{harm} (**Harmful CR**) across the four models we evaluated. As shown in Figure 3, the compliance rate on both harmful and harmless data increases as the λ decreases. This shows the ability to granularly adjust the boundary between safe and unsafe concepts by adjusting λ . For GEMMA-7B-IT, LLAMA-3-8B-INS and LLAMA2-7B-CHAT model, the harmful query compliance rate increases faster than the model of LLAMA2-13B-CHAT. This means the false refusal vectors are closer to the true refusal vectors in these two models, which affects more the true refusal behavior after the ablation. In comparison, the compliance rate on harmful data remains low when the λ decreases for LLAMA2-13B-CHAT, while the compliance rate on pseudo-harmful data increases substantially. This indicates that false refusal and true refusal vectors we extracted from LLAMA2-13B-CHAT are well separated and disentangled in the beginning before the orthogonalization. Therefore, changing the λ values has less impact on the true refusal.

The compliance rate curves give us an idea of how well the false refusal and true refusal are separated for the model and how to calibrate it. For models such as GEMMA-7B-IT, LLAMA2-7B-CHAT and LLAMA-3-8B-INS, staying in the range between 0.8 and 1.0 keeps the model relatively safe on harmful queries. Setting λ as 0.8 makes the model substantially more helpful, which proves a good trade-off between ‘harmlessness’ and ‘helpfulness’. As for LLAMA2-13B-CHAT, it is safer to further lower the λ to maximize the helpfulness of the model. One possible reason for such clear separation between false refusal and true refusal vectors is that LLAMA2-13B-CHAT has undergone heavy safety finetuning which leads to the lowest compliance rate on pseudo-harmful data, reaching 5.5% on **ORB-H**. This sets the boundary between a safe and unsafe query to an extreme level. Therefore, lowering the threshold for refusal will first affect the pseudo-harmful ones before affecting refusal to harmful queries, which have stronger harmful-related features.

To understand λ 's impact on general model capability, we also evaluate the model accuracy on MMLU in a 5-shot setting. As shown in Figure 3, the MMLU accuracy is not affected by changing λ across different models. The fact that performance is not affected is because we filtered the candidate refusal vectors that have largely shifted the first token probability distribution before the ablation.

5.3 REFUSAL VECTOR DISTRIBUTION

To better understand how the candidate true refusal and false refusal vectors are distributed in the transformer layers, we plot the refusal score changes across the token positions and layers when ablating the candidate refusal vectors in Figure 4. The y-axis is the relative token positions which are the positions of the post-instruction tokens such as `[ /INST ]`. The top row shows the refusal score changes when ablating the candidate true refusal vectors. From the second row, we plot the refusal score changes for the candidate false refusal vectors with increasing λ value. Compared with the true refusal vector distribution, the false refusal vectors generally have a lower refusal score change when ablated. This is because the false refusal vectors are extracted from pseudo-harmful data, which is less likely to induce refusal compared to the true refusal vectors. We also see a position overlap between the two kinds of vectors, as shown in the first two rows in the figure. This indicates that the false and true refusal is activated at similar positions and simply removing one could affect the other one, which is shown in our earlier experiment in Table 1. As we increase λ , the overall refusal score changes less, thus resulting in a weaker refusal removal effect. Furthermore, the distribution overlap between the candidate true refusal and false refusal vectors shrinks with increasing λ . For example, the most effective candidate true refusal vectors of LLAMA-3-8B-INS are concentrated at layer 13, while the most effective candidate false refusal vectors are mostly at layer 24 as λ increases from 0 to 0.6. Since we only select the most effective candidate refusal vector for the ablation, the final false refusal vector will be selected at different layer and token positions than the true refusal vector as we increase the λ , avoiding harming the true refusal behavior.

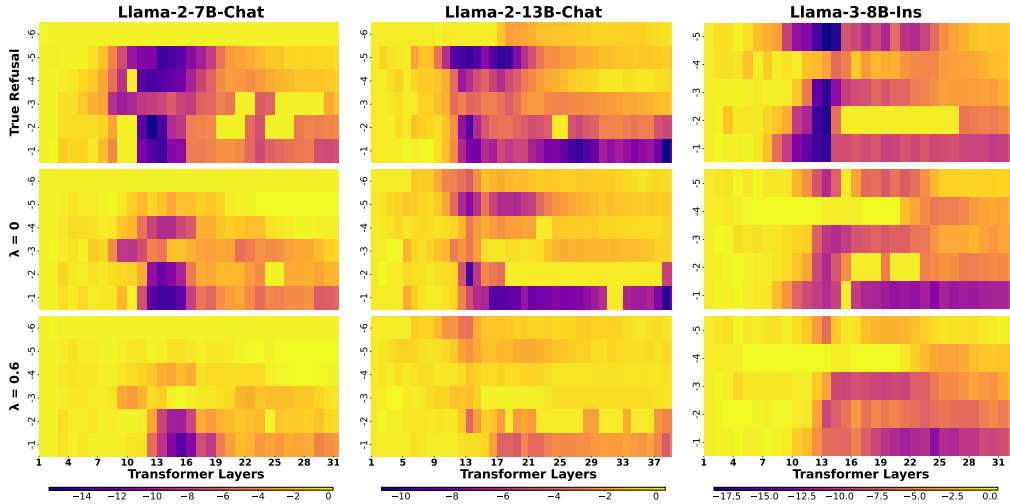


Figure 4: Refusal score changes when ablating the true (row 1) and false (row 2,3) refusal vectors extracted at certain layers and token positions. By increasing the value of λ , the refusal vectors have less impact on the model refusal behaviour.

5.4 DOES ADDING TRUE REFUSAL VECTOR HELP KEEPING MODEL SAFE?

Arditi et al. (2024) show that the model can be made to refuse more by adding the refusal vector to the activation stream. It is potentially beneficial to apply both the false refusal vector ablation and true refusal vector addition to get a more useful model without harming the safety guard. To investigate the effectiveness of this approach, we evaluate the performance of LLAMA-2-7B-CHAT on harmful data, pseudo-harmful data and general tasks under each combination of the adding coefficient α and orthogonalization coefficient λ . Results are given in Figure 9. We refer readers to

Appendix E for results of GEMMA-7B model. The baseline value below each plot shows the original model performance before the intervention. As shown in Figure 9, adding the true refusal vector can indeed make the model safer by inducing more refusal behavior. In the region where $\alpha \geq 0.2$ and $\lambda \geq 0.6$ as highlighted in the figure, the model behaves relatively safe on Jailbreakbench with a low compliance rate compared to the original model (baseline). This also leads to a low compliance rate on pseudo-harmful data such as OR-Bench-Hard. However, we see a degradation of general capacity measured by the accuracy on ARC-C, MMLU and perplexity on Wikitext. Therefore, the ablation operation is a more surgical approach than the addition, which was used by SCAN (An et al., 2024a) leading to general performance degradation, while our method does not.

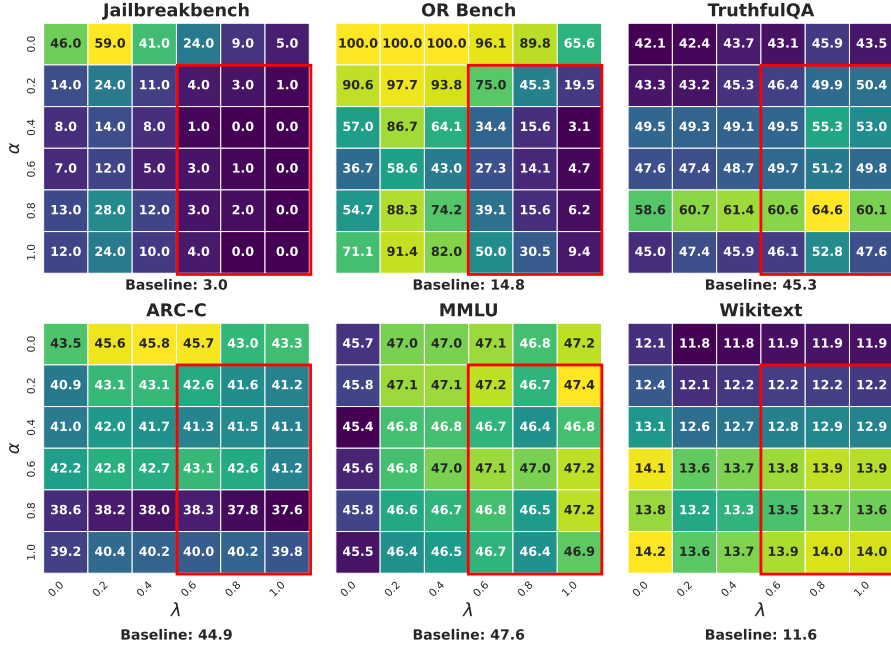


Figure 5: Performance of LLAMA2-7B-CHAT under different combination of α and λ . Highlighted area is where the modified model behaves relatively safe on Jailbreakbench compared to the original model (baseline). Vector addition improves the model safety by sacrificing the performance on ARC-C, MMLU and Wikitext.

However, the performance on TruthfulQA (Lin et al., 2022) increases as α increases. The BLUE score reaches 64.6 under the setting of $\alpha = 0.8$, $\lambda = 0.8$, compared to the baseline value of 45.3. As we inspect the model responses, adding the refusal vector tends to make the model more conservative and more likely to refuse to answer, which is favoured by the TruthfulQA questions. Based on the above observation, it is important to pay attention to the general performance degradation when adding true refusal vector to make the model safer, since it is not as surgical as the ablation operation. Therefore, we propose ablation as a mitigation approach versus the addition in Shi et al. (2024).

6 RELATED WORK

6.1 LANGUAGE MODEL SAFETY

LLM safety research aims to prevent LLMs from producing content that may harm individuals and society. The detection and elimination of undesirable attributes such as hate speech from LLM-generated texts has been extensively studied (Xu et al., 2020; Röttger et al., 2021; Sun et al., 2022; Vidgen et al., 2023). Recently, researchers are paying more attention to preventing LLMs from responding to malicious queries through alignment methods such as supervised fine-tuning (Bianchi et al., 2023) and RLHF (Bai et al., 2022a;b). In addition, efforts have been made to evaluate the safety of LLMs, ranging from near-term risks (Lin et al., 2022; Wang et al., 2023; Xie et al., 2024) to longer-term catastrophic risk potential (Hendrycks et al., 2023; Phuong et al., 2024).